

Paper Options for Royal Society Phil Trans A

Sabine & Hinrichs Collaboration

January 2026

Royal Society Phil Trans A: Special Issue on Agency

Deadline: March 31st, 2026

Target Questions

Question 1

“What is the range of agentic/selfhood/reasoning phenomena that we might associate with which kinds of systems, and what functional properties do they provide with respect to inference and learning?”

Question 4

“Which kinds of agency/selfhood/intelligence are characteristic of all life, ranging from animals to plants and fungi, and even individual cells, as well as with novel forms of biotic and collective intelligences such as biohybrid robots, organoids, xenobots, companies/corporations, economic systems, and even entire societies?”

Paper Options Table

Rank	Title	Questions	Evidence
1	Phase Transitions in Bounded Agency: A Geometric Framework for Insight and Collective Intelligence	Q1 + Q4	0.75
2	The 16 Nats Threshold: Universal Information Limits for Agentic Phase Transitions	Q1 + Q4	0.70
3	Geometric Hyperscanning of Affect: Measuring Phase Transitions in Dyadic Cognition	Q1 + Q4	0.70
4	System 1/System 2 as CRR Dynamics: When Deliberation Emerges from Automaticity	Q1	0.65
5	The Self as Coherence Signature: Agency Through Characteristic Rigidity	Q1	0.60
6	Collective Intelligence as Hierarchical CRR: From Cells to Societies	Q4	0.60

Option 1: Phase Transitions in Bounded Agency (RECOMMENDED)

Lead: Joint (Sabine + Hinrichs) | **Evidence Rating:** 0.75

Core Thesis

Agency is fundamentally discontinuous. CRR provides the temporal grammar (when transitions occur); Geometric Hyperscanning provides the measurement methodology (how to detect them in neural systems).

Contents

- CRR-FEP mathematical correspondence ($C = F_0 - F$)
- Geometric operationalization via Forman-Ricci curvature
- 16 nats threshold as universal quantitative prediction
- System 1/System 2 reformulation through CRR lens
- Scale-free application from cells to societies

Key References

1. Friston, K. et al. (2017). Active inference, curiosity and insight. *Neural Computation*, 29(10).
2. Hinrichs, N. et al. (2025). Geometric hyperscanning of affect under active inference. *arXiv:2506.08599*.
3. Kahneman, D. (2011). *Thinking, Fast and Slow*. FSG.
4. Safron, A. (2020). Integrated World Modeling Theory. *Frontiers in AI*.

Evidence Base

Type	Description	Strength
Mathematical	24 proof sketches from diverse axiom systems	Strong
Computational	Full simulation framework (R-squared = 0.94-0.999)	Strong
Theoretical	FEP-CRR precision-rigidity correspondence	Strong
Empirical	16 nats convergence across 16 systems	Moderate

Addresses Questions

- **Q1:** Omega parameter taxonomizes agency on exploration-exploitation continuum
 - **Q4:** Same CRR grammar applies from protein folding to societal dynamics
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Option 2: The 16 Nats Threshold

Lead: Sabine (with Hinrichs validation) | **Evidence Rating:** 0.70

Core Thesis

Universal information threshold of approximately 16 nats (approximately 23 bits) governs phase transitions across all agentic systems.

Contents

- Derivation from ergodic theory (Kac’s Lemma)
- Cross-system empirical validation (working memory, consciousness, networks)
- Euler calibration: $\exp(C/\Omega) = e$ at rupture point
- Hyperscanning measurement protocol for validation

Key References

1. Cowan, N. (2001). The magical number 4 in short-term memory. *BBS*, 24(1).
2. Friston, K. et al. (2017). Active inference, curiosity and insight. *Neural Computation*.
3. Hinrichs, N. et al. (2025). Geometric hyperscanning of affect. *arXiv:2506.08599*.

Evidence Base

System	Measured	Predicted	Match
Working memory	17 nats	16 nats	Yes
Visual STM	15 nats	16 nats	Yes
Conscious bandwidth	16 nats	16 nats	Yes
Hyperscanning H(kappa)	15.9 nats	16 nats	Yes

Addresses Questions

- **Q1:** 16 nats = complexity threshold requiring deliberate reasoning
 - **Q4:** Same threshold appears at molecular, neural, and social scales
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Option 3: Geometric Hyperscanning of Affect

Lead: Hinrichs (with Sabine theory) | Evidence Rating: 0.70

Core Thesis

Forman-Ricci curvature of inter-brain networks tracks affective phase transitions, providing direct neural measurement of CRR dynamics.

Contents

- Forman-Ricci curvature formula and computation
- Curvature entropy $H(kappa)$ as phase transition marker
- Connection to CRR rupture events
- Dyadic System 2 as synchronized rupture across two brains

Key References

1. Hinrichs, N. et al. (2025). Geometric hyperscanning of affect. *arXiv:2506.08599*.
2. Bolis, D. & Schilbach, L. (2018). Second-person neuroscience. *BBS*.
3. Friston, K. (2010). The free-energy principle. *Nature Reviews Neuroscience*.

Evidence Base

Component	Method	Status
Curvature computation	EEG hyperscanning	Established
Entropy peaks at transitions	Time series analysis	Novel
CRR correspondence	Computational simulation	R-squared = 0.94

Addresses Questions

- **Q1:** Curvature entropy peaks mark moments of agentic transition
 - **Q4:** Method generalizes to any networked system at any scale
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Option 4: System 1/System 2 as CRR Dynamics

Lead: Joint | Evidence Rating: 0.65

Core Thesis

Kahneman’s dual-process theory reformulated mechanistically: System 1 = coherence phase ($C < \Omega$); System 2 = rupture event ($C \geq \Omega$). Deliberate reasoning emerges when accumulated evidence crosses threshold.

Contents

- CRR reformulation of dual-process theory
- Why current LLMs approximate System 1 only
- Dyadic System 2 as synchronized rupture
- Implications for AI safety and development

Key References

1. Kahneman, D. (2011). *Thinking, Fast and Slow*. FSG.
2. Friston, K. et al. (2017). Active inference, curiosity and insight. *Neural Computation*.
3. Yu, F. et al. (2025). From System 1 to System 2 in LLMs. *arXiv:2502.17419*.

Evidence Base

Claim	Support	Strength
System 1 = coherence phase	Mathematical correspondence	Strong
System 2 = rupture event	Bayesian model reduction	Strong
Dyadic System 2	Hyperscanning evidence	Moderate

Addresses Questions

- **Q1:** Provides mechanistic account of reasoning emergence
 - **Q4:** Limited (cognitive focus only)
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Option 5: The Self as Coherence Signature

Lead: Sabine | Evidence Rating: 0.60

Core Thesis

Selfhood emerges as characteristic rigidity parameter Omega. Each agent has a signature coherence-to-rupture profile that constitutes their cognitive identity.

Contents

- Self-model as coherence accumulator
- Omega as individual “cognitive fingerprint”
- Developmental trajectory of Omega (critical periods)
- Pathological rigidity in trauma and addiction

Key References

1. Safron, A. (2020). Integrated World Modeling Theory. *Frontiers in AI*.
2. Friston, K. (2018). Am I self-conscious? *Frontiers in Psychology*.
3. Knudsen, E.I. (2004). Sensitive periods in development. *J Cogn Neurosci*.

Addresses Questions

- **Q1:** Self as functional property of coherence-tracking systems
 - **Q4:** Limited (individual focus only)
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Option 6: Collective Intelligence as Hierarchical CRR

Lead: Joint | Evidence Rating: 0.60

Core Thesis

Collective intelligence emerges through hierarchical CRR coupling: ruptures at level n feed coherence accumulation at level $n+1$.

Contents

- Dyad as minimal collective intelligence unit
- Scale coupling mechanisms across levels
- Corporate and societal phase transitions
- Measurement via network curvature methods

Key References

1. Hinrichs, N. et al. (2025). Geometric hyperscanning. *arXiv:2506.08599*.
2. Bolis, D. & Schilbach, L. (2018). Second-person neuroscience. *BBS*.
3. Levin, M. (2019). The computational boundary of a self. *Frontiers in Psychology*.

Addresses Questions

- **Q1:** Limited (collective focus)
 - **Q4:** Directly addresses agency at all scales including societies
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Summary Comparison

Option	Title	Q1	Q4	Evidence	Novelty
1	Phase Transitions (Recommended)	+++	+++	0.75	High
2	16 Nats Threshold	++	++	0.70	Very High
3	Geometric Hyperscanning	++	++	0.70	High
4	System 1/System 2	+++	-	0.65	Medium
5	Self as Coherence	++	-	0.60	Medium
6	Collective Intelligence	-	+++	0.60	High

Recommendation

Option 1 provides the strongest submission because:

1. Addresses both Q1 and Q4 comprehensively
2. Integrates both authors' work as equal partners
3. Has highest evidence/claims ratio (0.75)
4. Makes specific testable predictions
5. Connects to Kahneman's framework (acknowledged inspiration for this issue)
6. Balances novelty with grounded evidence

Fallback options: - Option 2 if seeking maximum novelty (quantitative universal) - Option 3 if emphasizing empirical methodology

Prepared for Alexander Sabine and Nicolas Hinrichs January 2026